Difference Equations

Dynamical systems in ecology, economics and engineering

Goal : To study the rate of change of various quantities over time.

- Population of various species of animals (e.g., coyotes and squirrels).
- Population of cities, countries, etc (e.g., people migrate to Canada and leave Canada).
- Market share of companies providing the same service (e.g., number of people subscribing to a service provided by a company).

The results of such processes are provided as a list of numbers produced at regular time intervals.

- Wildlife study (tag/release).
- Annual census.
- Monthly consumer data releases.

Example : Internet providers

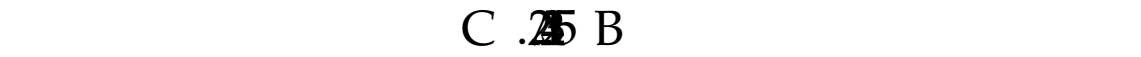
- In a country, three major companies provide high-speed internet : A, B, C.
- Most of the users subscribe to one of these companies (the market share of local companies is negligible).

A	B	C
.35	.15	.5

Market shares in the beginning of 2009 :the table shows the percentage of customers who subscribed to a given company in January 2009.

- Each year, a portion of customers of one company decide to switch to a different company.

C .25 B



How to interpret the arrows and numbers?

By the end of 2009 :

- A has lost 30% of its customers to B and 40% of its customers to C. (Therefore A retains 30% of its own customers.)
- B has lost 20% of its customers to A and 10% of its customers to C. (Therefore B retains 70% of its own customers.)
- C has lost 25% of its customers to A, and 20% of its customers to B. (Therefore C retains 55% of its own customers.)

C .25 B

We can summarize the above data in the following table :

From...	**A**	**B**	**C**	To...
	.3	.2	.25	**A**
	.3	.7	.2	**B**
	.4	.1	.55	**C**

$$\rightsquigarrow M = \begin{bmatrix} .3 & .2 & .25 \\ .3 & .7 & .2 \\ .4 & .1 & .55 \end{bmatrix}$$

Remark. The sum of entries in every column of M is equal to 1. Why?

Because the sum of every column is 100%.

Problem. After one year, what will be the market share of each company?

Solution.
The initial (i.e., current) market shares can be represented by a vector :

A	B	C
.35	.15	.5

$$\rightsquigarrow \quad \mathbf{x}_0 = \begin{bmatrix} A_0 \\ B_0 \\ C_0 \end{bmatrix} = \begin{bmatrix} .35 \\ .15 \\ .5 \end{bmatrix}$$

Let A_1, B_1, C_1 denote the market share of the corresponding company after one year. Then :

$$M = \begin{bmatrix} .3 & .2 & .25 \\ .3 & .7 & .2 \\ .4 & .1 & .55 \end{bmatrix} \quad \rightsquigarrow \quad \begin{cases} A_1 = .3A_0 + .2B_0 + .25C_0 \\ B_1 = .3A_0 + .7B_0 + .2C_0 \\ C_1 = .4A_0 + .1B_0 + .55C_0 \end{cases}$$

If we set $\mathbf{x}_1 = \begin{bmatrix} A_1 \\ B_1 \\ C_1 \end{bmatrix}$ then $\mathbf{x}_1 = M\mathbf{x}_0$

$$\mathbf{x}_1 = M\mathbf{x}_0 = \begin{bmatrix} .3 & .2 & .25 \\ .3 & .7 & .2 \\ .4 & .1 & .55 \end{bmatrix} \begin{bmatrix} .35 \\ .15 \\ .5 \end{bmatrix} = \begin{bmatrix} .26 \\ .31 \\ .43 \end{bmatrix}$$

Remark. The sum of entries of $\mathbf{x}_1$ is equal to 1 : .26+.31+.43=1
Why? Because they are the percentages of the market share.

You can use this fact to check your calculation.

Problem. What will be the market share of each company after two years?

Solution. If $\mathbf{x}_2 = \begin{bmatrix} A_2 \\ B_2 \\ C_2 \end{bmatrix}$ is the vector which shows the market share after two years, then again as before we have

$$\mathbf{x}_2 = M\mathbf{x}_1 = \begin{bmatrix} .3 & .2 & .25 \\ .3 & .7 & .2 \\ .4 & .1 & .55 \end{bmatrix} \begin{bmatrix} .26 \\ .31 \\ .43 \end{bmatrix} = \begin{bmatrix} .2475 \\ .381 \\ .3715 \end{bmatrix}$$

Note that $\mathbf{x}_2 = M\mathbf{x}_1 = M(M\mathbf{x}_0) = M^2\mathbf{x}_0$.

Remark. Again we have .2475+.381+.3715=1.

(You can use this fact to check your calculation.)

Problem. What will be the market share of each company after eight years?

(We are assuming that the migration rates of customers do not change.)

Solution. As before, we have :

$$\mathbf{x}_8 = M^8\mathbf{x}_0 = \begin{bmatrix} 0.24 \\ 0.448 \\ 0.312 \end{bmatrix}$$

Another example : population migration

Example

- A recent survey in Sudbury indicates that 5% of households which don't own a home will buy one by the end of the year. At the same time, 3% of households which own a home have decided to sell it by the end of the year.

Currently, homeowners constitute 80% of the population of Sudbury. What percentage of households will be home owners next year?

Solution. The current state can be represented by a vector :

$$\mathbf{x}_0 = \begin{bmatrix} .8 \\ .2 \end{bmatrix} \begin{matrix} \rightsquigarrow \text{ Home owners} \\ \rightsquigarrow \text{ Non-home owners} \end{matrix}$$

The migration matrix M is obtained as follows :

From...	Home owner	Non-home owner	To...
	.97	.05	Home owner
	.03	.95	Non-home owner

$$\rightsquigarrow M = \begin{bmatrix} .97 & .05 \\ .03 & .95 \end{bmatrix}$$

Next year's state is given by

$$\mathbf{x}_1 = M\mathbf{x}_0 = \begin{bmatrix} .97 & .05 \\ .03 & .95 \end{bmatrix} \begin{bmatrix} .8 \\ .2 \end{bmatrix} = \begin{bmatrix} 0.786 \\ 0.214 \end{bmatrix} \Rightarrow 78.6\% \text{ of households will be home owners.}$$

Example

What percentage of households will be home owners after 10 years? (Assume that the rates of buying and selling homes remain the same.)

Solution.

We have $\mathbf{x}_1 = M\mathbf{x}_0$, $\mathbf{x}_2 = M\mathbf{x}_1$, $\mathbf{x}_3 = M\mathbf{x}_2$, $\mathbf{x}_4 = \mathbf{x}_3$,...

$$\mathbf{x}_{10} = M^{10}\mathbf{x}_0 = \begin{bmatrix} 0.787 & 0.354 \\ 0.213 & 0.646 \end{bmatrix} \begin{bmatrix} .8 \\ .2 \end{bmatrix} = \begin{bmatrix} 0.7004 \\ 0.2996 \end{bmatrix}$$

which means that 70.04% of households will be home owners.

Terminology

A vector $\mathbf{x}$ is called a probability vector if :

- its entries are nonnegative,
- the sum of its entries is equal to 1.

Example. $\mathbf{x} = \begin{bmatrix} .3 \\ .2 \\ .4 \\ .1 \end{bmatrix}$ is a probability vector.

A square matrix P is called a stochastic matrix if its columns are probability vectors.

Example. $P = \begin{bmatrix} .3 & .4 & .75 \\ .2 & .15 & .1 \\ .5 & .45 & .15 \end{bmatrix}$ is a stocastic matrix.

Markov Chains

A Markov chain is a sequence of probability vectors

$$\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_p, \mathbf{x}_{p+1}, ...$$

together with a stochastic matrix M such that :

$$\mathbf{x}_{k+1} = M\mathbf{x}_k$$

The vectors $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, ...$ are called state vectors. The matrix M is called the migration matrix (or the transition matrix).

• Our next goal is to analyze the long term behavior of the Markov chain (i.e., to determine a vector $\mathbf{x}$ such that the vectors $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, ...$ converge to $\mathbf{x}$.)

Example. If $\mathbf{x}_0 = \begin{bmatrix} .23 \\ .77 \end{bmatrix}$, $\mathbf{x}_1 = \begin{bmatrix} .24 \\ .76 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} .245 \\ .755 \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} .2495 \\ .7505 \end{bmatrix}$, $\mathbf{x}_4 = \begin{bmatrix} .24995 \\ .75005 \end{bmatrix}$, ...then we can guess that the $\mathbf{x}_k$'s converge to $\mathbf{x} = \begin{bmatrix} .25 \\ .75 \end{bmatrix}$.

Example

The climate of the island of Nassau in the pacific ocean is somewhat unique: there is only one season, and days are either rainy or sunny.

Meteorological measurements have shown that :

- If a day is sunny, then there is a 70% chance that the next day is sunny too.
- If a day is rainy, then there is a 60% chance that the next day is rainy too.

If today (Tuesday) is sunny, what are the chances that it rains on Thursday?

Solution.
We can form the migration matrix as follows :

Today :	Rainy	Sunny	**Tomorrow :**
	.6	.3	Rainy
	.4	.7	Sunny

$$\rightsquigarrow M = \begin{bmatrix} .6 & .3 \\ .4 & .7 \end{bmatrix}$$ stochastic matrix

Today is sunny $\Rightarrow$ $\mathbf{x}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ (initial state)

$$\text{State of Thursday : } \mathbf{x}_2 = M^2\mathbf{x}_0 = \begin{bmatrix} .6 & .3 \\ .4 & .7 \end{bmatrix}^2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} .48 & .39 \\ .52 & .61 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} .39 \\ .61 \end{bmatrix}$$

$\Rightarrow$ There is a 61% chance that Thursday is sunny as well.

Example

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Meteorological measurements have shown that :

- If a day is sunny, then there is a 70% chance that the next day is sunny too.
- If a day is rainy, then there is a 60% chance that the next day is rainy too.

In the long run, what are the chances for a sunny day?

Solution. Suppose today is sunny. Then $\mathbf{x}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and we have :

$$\mathbf{x}_1 = M\mathbf{x}_0 = \begin{bmatrix} .3 \\ .7 \end{bmatrix} \qquad \mathbf{x}_2 = M^2\mathbf{x}_0 = \begin{bmatrix} .39 \\ .61 \end{bmatrix} \qquad \mathbf{x}_3 = M^3\mathbf{x}_0 = \begin{bmatrix} .417 \\ .583 \end{bmatrix}$$

$$\mathbf{x}_4 = M^4\mathbf{x}_0 = \begin{bmatrix} .4251 \\ .5749 \end{bmatrix} \qquad \mathbf{x}_5 = M^5\mathbf{x}_0 = \begin{bmatrix} .42753 \\ .57247 \end{bmatrix} \qquad \mathbf{x}_6 = M^6\mathbf{x}_0 = \begin{bmatrix} .428259 \\ .571741 \end{bmatrix}$$

$$\dots$$

We can guess that in the long run, there is approximately 57% chance that we have a sunny day...

- But what is the *exact* value???

Regular Stochastic Matrix

A stochastic matrix P is called regular if for some $k \geq 1$, all of the entries of P^k are strictly positive.

Example. The matrix $P = \begin{bmatrix} .6 & .3 & .2 \\ 0 & .3 & .3 \\ .4 & .4 & .5 \end{bmatrix}$ is a regular stochastic matrix because

$P^2 = \begin{bmatrix} 0.44 & 0.35 & 0.31 \\ 0.12 & 0.21 & 0.24 \\ 0.44 & 0.44 & 0.45 \end{bmatrix}$ and all of the entries of P^2 are strictly positive.

Steady-state vectors

Steady-state vector

If M is an $n \times n$ stochastic matrix, then a nonzero vector $\mathbf{x}$ in $\mathbb{R}^n$ is called a steady-state vector or an equilibrium vector if it has the following two properties :

- $\mathbf{x}$ is a probability vector, i.e., its entries are non-negative and the sum of entries of $\mathbf{x}$ equals 1.
- $M\mathbf{x} = \mathbf{x}$, i.e., $\mathbf{x}$ is an eigenvector of M corresponding to eigenvalue $\lambda = 1$.

Example. If $M = \begin{bmatrix} .6 & .3 \\ .4 & .7 \end{bmatrix}$ then $\mathbf{x} = \begin{bmatrix} \frac{3}{7} \\ \frac{4}{7} \end{bmatrix}$ is the steady-state vector of M because it is a probability vector and $M\mathbf{x} = \begin{bmatrix} .6 & .3 \\ .4 & .7 \end{bmatrix} \begin{bmatrix} \frac{3}{7} \\ \frac{4}{7} \end{bmatrix} = \begin{bmatrix} \frac{3}{7} \\ \frac{4}{7} \end{bmatrix} = \mathbf{x}$.

Existence of steady-state vectors

Theorem

A regular stochastic matrix M has a unique steady-state vector $\mathbf{x}$.

Moreover, if

$$\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \ldots$$

is a Markov chain with initial state $\mathbf{x}_0$ and migration matrix M (i.e., $\mathbf{x}_k = M\mathbf{x}_{k-1}$), then the $\mathbf{x}_i$'s converge to $\mathbf{x}$.

That is, as k gets bigger and bigger, the entries of $\mathbf{x}_k$ get closer and closer to the entries of $\mathbf{x}$.

If the migration matrix of a Markov chain is *regular stochastic*, then...

- the long-term behaviour of the Markov Chain is described by the steady-state vector.
- The eigenspace of M corresponding to the eigenvalue $\lambda = 1$ is one-dimensional.
- The steady-state vector $\mathbf{x}$ is the *unique* probability vector in the eigenspace of M corresponding to the eigenvalue $\lambda = 1$.
- The long-term behavior of Markov chain does not depend on the initial state (because it is always the steady-state vector).

Example

The climate of the island of Nassau in the pacific ocean is somewhat unique: there is only one season, and days are either rainy or sunny.

Meteorological measurements have shown that :

- If a day is sunny, then there is a 70% chance that the next day is sunny too.
- If a day is rainy, then there is a 60% chance that the next day is rainy too.

In the long run, what are the chances for a sunny day?

Solution. The migration matrix is $M = \begin{bmatrix} .6 & .3 \\ .4 & .7 \end{bmatrix}$, which is **regular stochastic**.

We compute the steady-state vector:

$$M - \mathrm{I} = \begin{bmatrix} .6 & .3 \\ .4 & .7 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -.4 & .3 \\ .4 & -.3 \end{bmatrix}$$

We solve $(M - \mathrm{I})\mathbf{x} = \mathbf{0}$ where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.

$$\left[\begin{array}{cc|c} -.4 & .3 & 0 \\ .4 & -.3 & 0 \end{array}\right] \xrightarrow[R_2 \to 10R_2]{R_1 \to 10R_1} \left[\begin{array}{cc|c} -4 & 3 & 0 \\ 4 & -3 & 0 \end{array}\right] \xrightarrow{R_1 \to -\frac{1}{4}R_1} \left[\begin{array}{cc|c} 1 & -\frac{3}{4} & 0 \\ 4 & -3 & 0 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 - 4R_1} \left[\begin{array}{cc|c} 1 & -\frac{3}{4} & 0 \\ 0 & 0 & 0 \end{array}\right] \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{3}{4}x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} \frac{3}{4} \\ 1 \end{bmatrix} \Rightarrow \boxed{\text{Basis: } \left\{ \begin{bmatrix} \frac{3}{4} \\ 1 \end{bmatrix} \right\}}$$

Example

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In the long run, what are the chances for a sunny day?

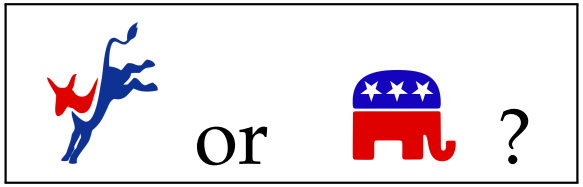
Basis: $\left\{ \begin{bmatrix} \frac{3}{4} \\ 1 \end{bmatrix} \right\}$

• You can also check your answer: $\begin{bmatrix} .6 & .3 \\ .4 & .7 \end{bmatrix} \begin{bmatrix} \frac{3}{4} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{3}{4} \\ 1 \end{bmatrix}$ ✓

But we need a probability vector, i.e. we need to find a t such that $t \begin{bmatrix} \frac{3}{4} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{3}{4}t \\ t \end{bmatrix}$ is a probability vector.

$$\frac{3}{4}t + t = 1 \;\Rightarrow\; t = \frac{4}{7} \;\Rightarrow \begin{bmatrix} \frac{3}{4}t \\ t \end{bmatrix} = \begin{bmatrix} \frac{3}{7} \\ \frac{4}{7} \end{bmatrix} \rightsquigarrow \text{steady-state vector}$$

$\Rightarrow$ In the long run, chances of having a sunny day is $\frac{4}{7} = .571428571\cdots$, i.e., about 57.14%



Example

In a country, 10% of those who vote Democratic in one election vote Republican in the next election, and 5% of those who vote Republican in one election vote Democratic in the next election.

In the last election, 55% of the voters voted Democratic. What percent of the voters are predicted to vote Democratic in the next election?

(We assume that essentially the same group of people vote in both elections.)

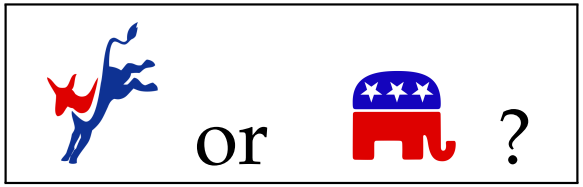
Solution.

The migration matrix for the election is $M = \begin{bmatrix} .9 & .05 \\ .1 & .95 \end{bmatrix}$. It is **regular stochastic.**

Initial state : $\mathbf{x}_0 = \begin{bmatrix} .55 \\ .45 \end{bmatrix}$

For the next election, we have $\mathbf{x}_1 = M\mathbf{x}_0 = \begin{bmatrix} .9 & .05 \\ .1 & .95 \end{bmatrix}\begin{bmatrix} .55 \\ .45 \end{bmatrix} = \begin{bmatrix} 0.5175 \\ 0.4825 \end{bmatrix}$

$\Rightarrow$ In the next election, 51.75% of voters will vote decomcratic.



Example

In a country, 10% of those who vote Democratic in one election vote Republican in the next election, and 5% of those who vote Republication in one election vote Democratic in the next election.

In the last election, 55% of the voters voted Democratic. What percent of the voters are predicted to vote Democratic in the election after the next election?

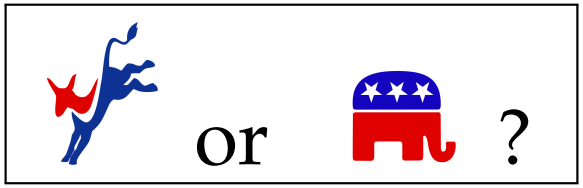
(We assume that essentially the same group of people vote in both elections.)

Solution.

The migration matrix for the election is $M = \begin{bmatrix} .9 & .05 \\ .1 & .95 \end{bmatrix}$.

$$\mathbf{x}_0 = \begin{bmatrix} .55 \\ .45 \end{bmatrix} \quad \Rightarrow \mathbf{x}_2 = M^2\mathbf{x}_0 = M\mathbf{x}_1 = \begin{bmatrix} .9 & .05 \\ .1 & .95 \end{bmatrix} \begin{bmatrix} 0.5175 \\ 0.4825 \end{bmatrix} = \begin{bmatrix} 0.489875 \\ 0.510125 \end{bmatrix}$$

48.98% of voters will vote Democratic.



Example

In a country, 10% of those who vote Democratic in one election vote Republican in the next election, and 5% of those who vote Republication in one election vote Democratic in the next election.

In the last election, 55% of the voters voted Democratic. In the long run, what percent of votes are predicted to go to democrats?

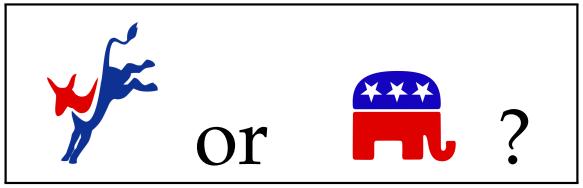
(We assume that essentially the same group of people vote in both elections.)

Solution.

The migration matrix for the election is $M = \begin{bmatrix} .9 & .05 \\ .1 & .95 \end{bmatrix}$, which is a **regular stochastic** matrix. Next we find the steady-state vector:

$$(M - \mathrm{I})\mathbf{x} = \mathbf{0} \quad M - \mathrm{I} = \begin{bmatrix} .9 & .05 \\ .1 & .95 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -.1 & .05 \\ .1 & -.05 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} -.1 & .05 & 0 \\ .1 & -.05 & 0 \end{array}\right] \xrightarrow[R_2 \to 10R_2]{R_1 \to -10R_1} \left[\begin{array}{cc|c} 1 & -\frac{1}{2} & 0 \\ 1 & -\frac{1}{2} & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{cc|c} 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{array}\right]$$



Example

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In the last election, 55% of the voters voted Democratic. In the long run, what percent of votes are predicted to go to democrats?

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$$\left[\begin{array}{cc|c} -.1 & .05 & 0 \\ .1 & -.05 & 0 \end{array}\right] \xrightarrow[R_2 \to 10R_2]{R_1 \to -10R_1} \left[\begin{array}{cc|c} 1 & -\frac{1}{2} & 0 \\ 1 & -\frac{1}{2} & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{cc|c} 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix} \Rightarrow \boxed{\text{Basis for the eigenspace : } \left\{ \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix} \right\}}$$

Probability vector: $t \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}t \\ t \end{bmatrix}$ $\frac{1}{2}t + t = 1 \Rightarrow t = \frac{2}{3} \Rightarrow t \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}$

$\Rightarrow$ In the long run, $\frac{1}{3}$ (i.e., approximately 33.33%) of the voters will vote Democratic.

Revisiting the example of internet providers

- In a country, three major companies (called A, B, and C) provide high-speed internet.
- Most of the users subscribe to one of these companies (the market share of local companies is negligible).

A	B	C
.35	.15	.5

Market shares in the beginning of 2009 :the table shows the percentage of customers which subscribed to a given company in January 2009.

- Each year, a portion of customers of one company decide to switch to a different company.

From...	**A**	**B**	**C**	To...
	.3	.2	.25	**A**
	.3	.7	.2	**B**
	.4	.1	.55	**C**

$$\rightsquigarrow M = \begin{bmatrix} .3 & .2 & .25 \\ .3 & .7 & .2 \\ .4 & .1 & .55 \end{bmatrix}$$

Problem. In the long run, what will be the market share of each of the companies?

Solution. The migration matrix $M = \begin{bmatrix} .3 & .2 & .25 \\ .3 & .7 & .2 \\ .4 & .1 & .55 \end{bmatrix}$ is a regular stochastic matrix. We find the steady-state vector of M.

We solve the equation $(M - \mathrm{I})\mathbf{x} = \mathbf{0}$ for $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$$M - \mathrm{I} = \begin{bmatrix} .3 & .2 & .25 \\ .3 & .7 & .2 \\ .4 & .1 & .55 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -.7 & .2 & .25 \\ .3 & -.3 & .2 \\ .4 & .1 & -.45 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -.7 & .2 & .25 & 0 \\ .3 & -.3 & .2 & 0 \\ .4 & .1 & -.45 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & 0 & -\frac{23}{30} & 0 \\ 0 & 1 & -\frac{43}{30} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

x_1, x_2 : basic, x_3 : free.

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{23}{30}x_3 \\ \frac{43}{30}x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} \frac{23}{30} \\ \frac{43}{30} \\ 1 \end{bmatrix}$$

To find the steady-state vector :

$$t\begin{bmatrix}\frac{23}{30}\\ \frac{43}{30}\\ 1\end{bmatrix} = \begin{bmatrix}\frac{23}{30}t\\ \frac{43}{30}t\\ t\end{bmatrix}$$

$$\frac{23}{30}t + \frac{43}{30}t + t = 1 \;\Rightarrow\; 96t = 30 \;\Rightarrow\; t = \frac{30}{96}$$

$$\Rightarrow\; \mathbf{x} = \begin{bmatrix}\frac{23}{96}\\ \frac{43}{96}\\ \frac{30}{96}\end{bmatrix} \text{ is the steady-state vector.}$$

$\Rightarrow$ In the long run, the market shares of the companies are :

Company	Market share
A	$\frac{23}{96} \approx 23.95\%$
B	$\frac{43}{96} \approx 44.79\%$
C	$\frac{30}{96} \approx 31.25\%$.